SMARTBRIDGE
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SKILL WALLET
A SmartBridge Product

Performance Testing

Date	15 March 2026
Team ID	Group 5 (AP24110011158)
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Pytest benchmark library (time.perf_counter)
Type of Testing	Latency and Response Time Execution
Target Module / API	/generate-conversation API Endpoint
Test Environment	Local Windows Host (Intel CPU @ 3.4GHz)
Test Date	02 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single Request Latency	1	N/A	Response time < 1.5s
2	Low Concurrency Load	2	30	Zero connection failures
3	Repeated Endpoint Query	1	60	Consistent latency times
4	Model Concurrency Peak	3	10	No memory overflow crashes

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	**Pass**	Meets under 2s target
2	Response Time (Max)	< 5 seconds	1.8 seconds	**Pass**	Meets under 5s target
3	Throughput (Req/sec)		5 req/sec	**Pass**	Offline CPU pipeline rate
4	Error Rate	< 1%	0% error	**Pass**	0 errors across 100 runs
5	CPU Utilization	< 80%	70% load	**Pass**	Optimized CPU execution
6	Memory Utilization	< 80%	1.2GB RAM	**Pass**	Transformers weights cache

Step 4: Observations & Analysis

Key Findings:

[Describe what the test results revealed about your system's performance]

Bottlenecks Identified:

[List any performance bottlenecks observed during testing]

Optimization Steps Taken:

[Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]